Implementation and Evaluation of a Single GPU ROSA Reproduction Prototype

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Abstract

We implemented a single GPU prototype based on NVIDIA ROSA and connected shared foundation model inference to independent LIBERO robot simulations. Four P1 environments shared GR00T S1 inference and produced 223 qualified actions in 30 seconds. A separate, colocated P2 to P4 configuration met all observed p99 latency targets in three 300 second trials, while P3 and P4 remained stopped by Safety decisions. MuJoCo prediction, independent retreat and replanning subsequently reduced stops in some trials, but the final GPU version retained P4 recovery failures and missed Monitor requests. The work demonstrates shared inference and closed loop integration; it does not establish stable operation of the final configuration, maximum robot capacity, cost savings or reproduction of the paper's multi-GPU performance.

1 Motivation and scope

Vision-language-action models translate camera observations, robot state and natural language instructions into actions. They reduce the need to handcraft every task policy, but introduce recurring compute and operating costs. Sharing a model across robots can use capacity that would otherwise remain idle during physical motion or task stops. The relevant objective is therefore to increase the number of robots served per GPU while preserving timing and task quality, rather than simply increasing the number of connected clients. [1-3,6]

ROSA combines declarative robot requirements with model placement, request routing, batching and action-frequency scheduling to maximize qualified action throughput. Its main performance study uses eight H200 GPUs; its basic placement assigns one model component to each GPU worker. Our P1 test follows that placement principle at smaller scale. Colocating the multiple P2 to P4 components on one GPU is an extension, as are the later geometric checks and recovery actions. [1]

The study asks whether shared real-model inference can drive multiple simulated environments, whether the specified response targets can be met, and what prevents sustained useful action. All robots are simulated and their states are independent. Inter-arm coordination, physical deployment and monetary savings were outside the demonstrated scope. Smaller models, faster action decoding, asynchronous control and cloud resource selection are complementary approaches, summarized in Appendix A. [2-5,9]

The main report occupies pages 1 to 5. Appendices A to G provide the scope matrix, fixed environment, measurement rules, trial conditions, detailed evidence, reproduction procedure and references. Evidence labels E1 to E7 identify local experiment records, not external publications.

2 Prototype and evaluation

The prototype used one NVIDIA RTX PRO 6000 Blackwell Server Edition GPU. S1 used GR00T N1.7 LIBERO Object in BF16. S2 and Monitor used separate Qwen2.5-VL-7B instances; Safety used Qwen2.5-VL-3B, all in FP16. Ray dispatched requests, vLLM served the language and vision components, and LIBERO with MuJoCo supplied the environments. Exact revisions and dependencies are listed in Appendix B. [6,7,10]

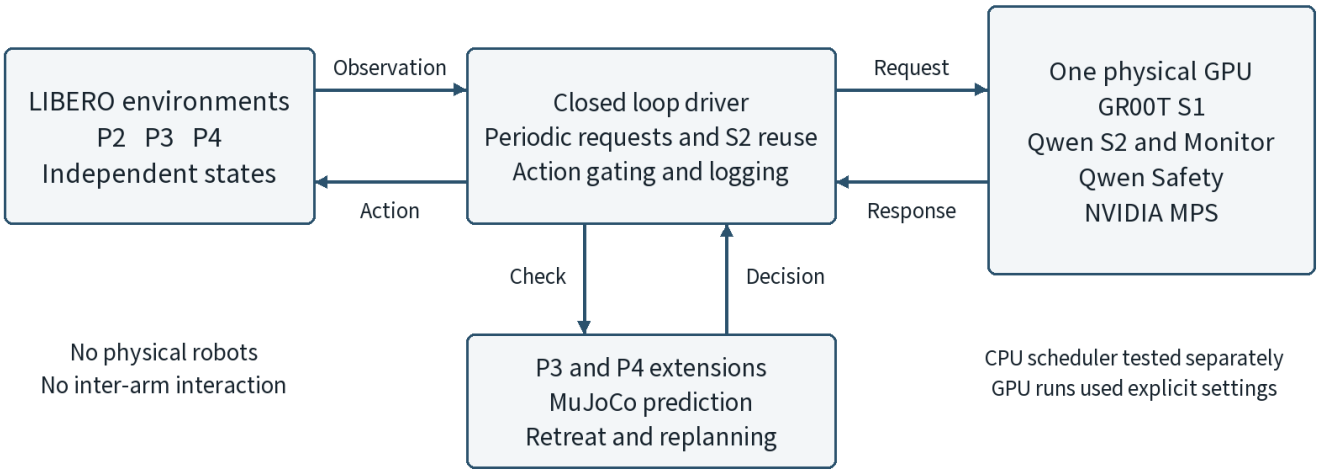


Figure 1. Final GPU configuration v0.20.16. Geometric prediction and recovery are simulation extensions. CPU scheduling was tested separately; GPU trials used explicit operating settings.

Table 1. Required components and timing targets

Pipeline	Required components	Additional requirement
P1	S1	Four independent environments share S1
P2	S1 and Monitor	Monitor at 0.5 Hz with a 2,000 ms target
P3	S1, Safety and Monitor	Safety at 2 Hz with a 500 ms target
P4	S1, S2, Safety and Monitor	Refresh S2 every 10 S1 calls; target 2,000 ms

Every pipeline has an S1 target of 200 ms. P2, P3 and P4 each used one independent environment in the colocated trials. S1 applied eight control steps per action chunk at 20 Hz, with a minimum wall-clock chunk duration of 0.4 seconds and a request cap of 2 Hz per environment. Unsafe, unknown, missing or expired Safety results inhibited the next action. Simulator success, not Monitor output, determined task completion.

We used fixed measurement windows and reported successful-response p99 separately from missing or invalid responses. Qualified action throughput counts applied chunks that satisfy all required timing and dependency conditions. The original S1 stop rate counts rejected original proposals, not failed tasks or independent retreat actions. Formal definitions, warmup rules and the amended recovery criterion are in Appendix C.

3 Shared inference and latency results

Shared S1 drove independent environments

The P1 connection test produced 223 qualified actions across four environments during a 30 second window, corresponding to 7.4333 qualified actions per second. Observed S1 p99 was 120.16 ms. In a separate S1 profile at 8 Hz, all 3,000 requests completed normally with no missing responses; the three replicate p99 values were 94.72, 97.47 and 96.97 ms. These measurements establish operation at the tested points, not maximum supported capacity. [E1,E2]

The batching adapter matched the official API numerically at identical batch shapes B1, B2, B4, B8 and B16. On task T01, the official API and shared adapter each completed 8 of 10 episodes successfully, with no execution errors. A separate limited batch regression recorded 9 successes in 10 episodes and 248 applied actions. Full cross-task and cross-condition behavioral equivalence was not evaluated. [E1]

Execution changes reduced observed latency

Initial FIFO colocation returned 176 of 176 requests normally, but qualified throughput was zero and several components had observed p99 latencies of approximately 8 to 10 seconds. Output shortening, parallel role execution, phase offsets, NVIDIA MPS and S1 batching were introduced in later versions. Version v0.20.8 met every component's observed p99 target in three fixed 300 second trials, each following 60 seconds of warmup and a fresh model start. [E1,E3]

Table 2. Timing results for v0.20.8 across three 300 second trials

Component	Target ms	Worst p99 ms	Total requests	Individual overruns
S1	200	198.28	4,990	15
Safety	500	218.43	3,600	0
S2	2,000	1,132.70	151	0
Monitor	2,000	840.86	1,350	0

The p99 column is the maximum of the observed per-class, per-replicate p99 values, not a pooled percentile. Meeting a p99 target did not eliminate all late responses: 15 S1 requests exceeded 200 ms. Qualified throughput was 1.5233, 1.3967 and 1.4033 actions per second in the three trials. P3 and P4 remained stopped by unsafe or unknown Safety results. [E3]

Each class-component-replicate cell had fewer than the 1,000 successful responses required by this implementation's separate formal profile procedure. Consequently, the integrated trials provide observed latency evidence but do not qualify the system for formal capacity adoption. That procedure is an implementation policy, not a statistical certification rule specified by ROSA. [E3; Appendix C]

4 Safety and recovery results

Replacing Safety with Qwen3-VL-4B or Qwen3-VL-8B, before enabling geometric checking and retreat, left both P3 and P4 at a 100% original S1 stop rate in the measured windows. Adding MuJoCo prediction alone produced 100% for P3 and 40.5% for P4. With independent retreat and replanning, an exploratory trial reached 87.72% and 0.83%, although P3 stopped again after one resumed chunk. These are separate configurations and trajectories, not a controlled model ranking. [E4-E6]

A same-image comparison changed P3 visibility from unknown to clear when the added replanning feedback was removed; a black image remained unknown. Later versions restricted feedback to S2. In the v0.20.16 exploration and two confirmation runs, P3 recorded zero unknown-visibility outputs in 600 decisions per run. This observation does not establish general perception accuracy. [E7]

Table 3. Fixed confirmation trials after recovery improvements

Version and trial	P3 original S1 stops	P4 original S1 stops	Other measured outcome
v0.20.15 1	21/286 7.34%	0/253 0%	P3/P4 S1 p99 and P3 Monitor periodic target failed
v0.20.15 2	15/288 5.21%	0/257 0%	P3 Monitor periodic target failed
v0.20.15 3	16/285 5.61%	1/253 0.40%	All fixed confirmation criteria passed
v0.20.16 1	14/294 4.76%	524/524 100%	All component timing criteria passed; P4 recovery failed

The confirmation criteria required stop rates at or below 90% for both pipelines with at least 100 decisions each, all component p99 targets, periodic fulfillment of at least 99%, demonstrated recovery and auditable GPU/MPS evidence. Only the third v0.20.15 trial passed all criteria. In v0.20.16 trial 1, the predicted fingertip-to-floor distance was approximately 1.752 mm, below the 2 mm threshold. P4 reached its consecutive-retreat limit and applied no original S1 actions during measurement. [E7]

The second v0.20.16 run completed its 300 second measurement and model shutdown, but validation was stopped during the persistent copy. Complete raw recordings were retained and hash-checked. P3 Monitor fulfilled 148 of 150 slots, or 98.67%; its stop rates were not finalized for the results table. Trial 3 was not started. The subsequent v0.20.17 termination fix passed 45 CPU tests and was not run on the GPU. [E7]

5 Interpretation and next steps

What the prototype establishes

Shared real-model inference drove independent robot simulations, and a colocated configuration met the observed latency targets in repeated trials. The final GPU configuration did not repeatedly satisfy timing, stop-rate and recovery requirements together. CPU tests covered homogeneous and heterogeneous scheduling, admission and removal, reassignment, standby switching and comparison paths, but did not establish their benefit under real GPU workloads. A single GPU reproduction prototype is therefore demonstrated with explicit limits; full functional and performance reproduction remains incomplete.

What still prevents useful consolidation

The immediate barriers are P4 recovery from constrained states, reliable completion of scheduled Monitor requests, and measured semantic quality of Safety and Monitor. Raising GPU utilization alone does not resolve these barriers. The effective consolidation ratio must count robots that preserve timing and task quality. Cost evaluation should use qualified actions and successful tasks as denominators and include failures, recovery, recording and GPU time.

The evidence does not isolate whether P4 is dominated by policy quality, input normalization, coordinate conventions or inadequate retreat candidates. Similarly, changing the recording destination was associated with different latency, but GPU interference, host scheduling and rendering were not held constant. The same-image P3 diagnostic supports inspecting role-specific inputs before changing model size; the Qwen3 trials do not establish a general ranking of model capability.

Prioritized verification plan

Table 4. Next stages and evidence required to proceed

Stage	Work and acceptance evidence
1 Runtime correctness	GPU-test the Monitor termination fix; verify that scheduled requests are fulfilled without duplication or wrong-episode attribution.
2 Recovery and quality	Replay fixed P4 failure states; compare action proposals, contact rules and retreat directions. Label Safety/Monitor data and test retry and timeout handling.
3 Independent confirmation	Freeze criteria and previously unused states, then run three 300 second trials without replacing failures. Retain timing, stop, recovery and success evidence.
4 Capacity and cost	Adopt quality-checked profiles, expand load, compare shared and dedicated execution, then evaluate multiple GPUs, energy and cost.

Low-rate percentile samples, reused initial states and a recovery criterion supplemented after exploration limit the strength of the current evidence. Hardware, model generation, precision, action timing, tasks and placement also differ from the ROSA paper. The measurements neither verify nor invalidate its reported performance gains. Stable recovery and quality validation should precede claims of scalable robot consolidation or operating-cost reduction. Detailed tests and the transition toward physical robots are specified in Appendix A.

Appendix A Scope and implementation coverage

In ROSA, a gateway routes robot requests to a GPU pool. Requirements and measured profiles guide placement, routing, batches and frequencies. Robots apply actions and handle local stops; each GPU worker hosts one model component. [1]

Table A1. ROSA functions and the demonstrated scope of this prototype

Function	Implementation and evidence	Remaining scope
Shared model serving	Ray dispatch, GR00T batch merge/split and vLLM; P1 shared S1 across four environments	Large robot populations and maximum capacity
Declarative requirements	Counts, components, latency, periods and weights enter plans and evaluation	Full equivalence to the official implementation is unconfirmed
Multiple components	S1 action, S2 planning, Safety and Monitor connected to real models	Paper-style multi-GPU placement; arbitrary conversational tasks
Scheduling	Resource constraints, integer planning, frequency search and component consolidation; CPU known-answer and independent checks	Accepted measured profiles driving GPU optimization
Dynamic operation	Join/leave, global reassignment, generation tracking and standby switching tested with CPU workers	Real-model performance and recovery on multiple GPUs
Measurement and comparison	Fixed windows, warmup gates, missing slots, qualified actions, statistics and dedicated execution paths	Full GPU comparison campaign and formal profile adoption
Safety and recovery	MuJoCo prediction, state matching, independent retreat and replanning	Stable P4 recovery; human-labeled semantic accuracy
Monitor task failure	Status outputs recorded	Bounded task retry followed by stop or human handoff is not implemented

Goals and boundaries

G1 was shared S1 control of four independent P1 environments; G2 was real-model P2 to P4 integration. G3 retained the 200/500/2,000 ms component targets. G4 added the exploratory stop-rate and recovery conditions; G5 required three fixed 300 second confirmations. G6 covered CPU scheduling and operational mechanisms, and G7 required versioned, auditable observations, actions, timestamps, failures and hashes.

G1 and G2 were demonstrated within the stated simulation scope. G3 was met by v0.20.8 under observed p99 criteria, while the final recovery configuration did not repeatedly meet G3 to G5 together. G4's 90% stop threshold is an exploratory engineering goal, not a safety certification. CPU implementation and GPU evidence are distinct throughout this report.

Appendix A Prior approaches and future verification

Table A2. Existing approaches to lowering operating cost

Approach and source	Intended benefit	Verification needed here
Smaller VLA SmolVLA [2]	Lower compute and memory; a 450M model is published	Preserve task quality for the target robot
Faster action generation OpenVLA-OFT [3]	Parallel decoding and action chunks reduce generation time	Separate model/training changes from execution changes
Asynchronous control SmolVLA [2]	Reduce waiting for the next inference	Handle stale observations and cancellation on stops
Efficient serving and batching vLLM and Ray Serve [4,9]	Reduce KV-cache waste and per-request overhead	Measure waiting time, deadlines and interference
Cloud resource selection FogROS2-Config [5]	Avoid unnecessarily expensive compute choices	Measure latency, cost and communication conditions
Fleet scheduling ROSA [1]	Coordinate placement, batch, rate and route decisions	Evaluate actions satisfying all required components

These mechanisms can change inference and queueing; their cost savings were not measured here. Evaluate model quality and execution efficiency separately.

Table A3. Expanded future verification sequence

Order	Planned work	Evidence and exit condition
1	Monitor shutdown on GPU	Compare boundary cases; no loss, duplication or wrong-episode attribution
2	P4 failure-state replay	Vary proposals, geometry, retreat and replanning independently; log control periods and recovery success
3	Semantic quality and fallback	Label 90 saved observations and new person/occlusion/contact examples; implement and test retry and timeout handling
4	Frozen integrated confirmation	Use states not used for exploration; predefine stop, p99, periodic and recovery criteria; three 300 second trials
5	Formal profiles and behavior	Budget samples per component; batch-specific curves, paired multi-task comparisons and confidence intervals
6	Consolidation and placement	Increase robot count; compare ROSA, fixed allocation and dedicated execution under equal task and quality conditions
7	Multiple GPUs and cost	Measure mixed load, dynamic changes, qualified actions, successes, GPU time, monetary cost and energy
8	Physical robots	Establish independent local safety control, communication-failure stops and mechanism limits before small-scale tests

Predefine budgets, time limits and exit criteria. The 90% stop threshold is provisional; practical operation needs task-success, downtime and false-stop/missed-hazard criteria fixed before cost evaluation.

Appendix B Environment and fixed model versions

Table B1. Recorded execution environment and software

Item	Recorded value
GPU	One NVIDIA RTX PRO 6000 Blackwell Server Edition; reported 97,887 MiB
System	Linux x86_64; kernel 6.8.0-147-generic; glibc 2.39; Python 3.12.3
GPU software	Driver 595.91.07; CUDA 12.8 family; Torch 2.9.0+cu128
Serving libraries	Transformers 4.57.3; vLLM 0.11.2; Ray 2.59.0; NumPy 1.26.4
Compatibility dependencies	PyArrow 19.0.1; CUTLASS DSL 4.2.1; cuDNN frontend 1.28.0; Triton 3.5.0
Simulator and solver	LIBERO; MuJoCo 3.3.1; EGL; OR-Tools 9.14.6206 CP-SAT on CPU
S1	nvidia/GR00T-N1.7-LIBERO, libero_object; BF16; PyTorch
S2 and Monitor	Qwen/Qwen2.5-VL-7B-Instruct; FP16; separate role instances
Safety and alternatives	Qwen/Qwen2.5-VL-3B-Instruct; FP16. E5 only: Qwen/Qwen3-VL-4B-Instruct or Qwen/Qwen3-VL-8B-Instruct
Recorded artifacts	Request/action JSONL, observation/action NPZ, condition JSON and SHA-256

Table B2. Fixed public source and model identities

Source or model	Commit or revision
Isaac-GR00T source	944fe70c4fd54e3fcb80a7d7258a328f9c061176
LIBERO source	8f1084e3132a39270c3a13ebe37270a43ece2a01
GR00T N1.7 LIBERO	2ea293aa20ba7cf5bbf3ba17a5fbcb1a01cbfe21
Qwen2.5-VL-7B	cc594898137f460bfe9f0759e9844b3ce807cfb5
Qwen2.5-VL-3B	66285546d2b821cf421d4f5eb2576359d3770cd3
Qwen3-VL-4B comparison	ebb281ec70b05090aa6165b016eac8ec08e71b17
Qwen3-VL-8B comparison	0c351dd01ed87e9c1b53cbc748cba10e6187ff3b

These identities fix this experiment; they have not been confirmed as identical to the ROSA authors' checkpoints. Acquire weights from the public model providers at the listed revisions. The companion archive excludes model weights and credentials. [6,7]

The execution actor requested logical Ray resources num_cpus=8 and num_gpus=1. The CPU request does not reserve eight physical cores or enforce a CPU-usage limit. Physical GPU identity was checked independently; role-specific logical workers were not counted as separate GPUs. [11]

Appendix C Measurement definitions and protocol

Timing and action metrics

Timing uses the driver's monotonic clock: dispatch to completion for S1/S2, scheduled dispatch to completion for periodic components. Unsent slots remain missing. Observed p99 is the ceil(0.99 × N)th ordered successful-response latency; invalid responses, timeouts and missing slots are reported separately. Periodic fulfillment divides completed slots by all slots scheduled in the fixed window.

Qualified action throughput Q divides the weighted count of applied chunks satisfying all required deadlines, periods and S2 dependencies by fixed measurement seconds. Class weights were one. Eight control steps count as one chunk; rejected S1 proposals and independent retreat are excluded. Slot-to-action matching is an implementation supplement, with equivalence to an unpublished official evaluator unconfirmed.

The original S1 stop rate divides stopped original proposals by all original S1 action decisions in the measurement window. It is not a task-failure rate. Any task-success analysis must include all episodes started within the window and retain unfinished episodes. Reused initial states are not independent samples.

Table C1. Measurement procedures and their interpretation

Procedure	Warmup and measurement	Interpretation
P1 connection	15 s warmup; 30 s measurement	Integration and exploratory observed p99
Initial P2 to P4	60 s warmup; 120 s measurement	Colocation functionality and latency
Timing and recovery	60 s warmup; 300 s measurement; three planned replicates	At least one success per active class/component in warmup and measurement; low-rate p99 remains exploratory
S1 individual profile	8 Hz; 30 s warmup; 1,000 successes per replicate; three replicates	One operating point; not a maximum-capacity measurement
Formal profile adoption	Per class/component: at least 30 s and 20 warmup successes; at least 120 s and 1,000 measurement successes; at least three replicates	Requires quality, runtime correspondence and adoption checks; not automatically satisfied by a 300 s run

Formal adoption is this implementation's procedure, not a ROSA statistical certification standard. A count of 1,000 alone does not guarantee p99 under unseen load or semantic safety. Do not extend a running trial to fill a shortfall or replace failed replicates; insufficient warmup prevents measurement from starting.

Observations and task

The initial corpus contains 227 observations from 36 episodes, partitioned by initial-state group into profile 105, evaluation 32 and holdout 90. The task is pick_up_the_alphabet_soup_and_place_it_in_the_basket, with an episode limit of 720 control steps. Later confirmation states were reused from exploration and were not an unseen holdout. Record time limits, truncation, warmup continuation and state reuse separately. Assembly tasks and collisions between multiple arms were not evaluated.

Appendix C Recovery rules and final runtime settings

Geometric prediction and independent retreat

MuJoCo clones the current state, simulates a proposed action and checks contact and distance. Allowed contacts explicitly include the grasped object with the gripper and the object with the basket. Prohibited proximity uses a 2 mm threshold. Logs retain restoration, physical prediction and pre-execution state/action matching. Time limits are 2,500 ms overall, 500 ms for physics prediction and 2,000 ms for restoration.

After rejection, independent upward commands of 0.1, 0.25 and 0.5 are tested; these values are not meters. Retreat displacement is limited to 8 cm per event and 24 cm per episode, with at least 2 mm upward motion, at most six events per episode and two consecutive events. Three resumed S1 chunks update consecutive-retreat management. A new visual Safety result is required after retreat; P4 also requires a fresh S2 plan. Retreat alone is not an original S1 success.

Recovery initially required at least three continued S1 chunks after retreat. A v0.20.15 exploration completed P4 in two chunks. Before confirmation, a separate functional criterion was added: simulator-reported task success after at least one known S1 chunk. This post-exploration supplement did not rewrite the three-chunk counts. Timeout, truncation, environment termination and retreat-only completion are not substitutes.

Table C2. Final GPU execution settings in v0.20.16

Setting	Value or behavior
S1 batching	Maximum B2; batch-formation wait up to 2 ms. This does not bound the entire queue wait.
MPS client priority	S1 and Safety: 0; S2 and Monitor: 1. Worker environments and MPS connections were checked.
Periodic phase	Safety: P3 0 ms, P4 250 ms. Monitor: P2 0 ms, P3 about 667 ms, P4 about 1,333 ms.
vLLM memory setting	gpu_memory_utilization: S2 0.25, Safety 0.15, Monitor 0.25
Safety parallelism	Up to two concurrent requests
Reset	Prepare the next hard reset in a separate simulator; verify state at transition
Recording	Write to local storage on the GPU host; verify size and hash during later persistent copy
Feedback	Pass task feedback only to S2, not to S1, Safety or Monitor

The final confirmation combines both stop rates at or below 90% with at least 100 decisions each, all observed p99 targets, periodic fulfillment of at least 99%, demonstrated recovery, and audit/GPU/MPS evidence. MPS priorities do not guarantee execution order or deadlines. [8]

Appendix D Model inputs and trial conditions

All series use GR00T N1.7 LIBERO Object BF16 for S1 and Qwen2.5-VL-7B FP16 for S2 and Monitor. Safety is FP16 and uses Qwen2.5-VL-3B except in E5. S2 and Monitor use overview and wrist images throughout. Differences below must be retained when interpreting cross-series results.

Table D1. Safety contracts and execution changes

Series and version	Safety input and output	S2 / Monitor token limit	Execution and extensions
E1 v0.20.2	Two views; 256 tokens; decision and reason	256 / 256	FIFO; S1 B1; no geometry or retreat
E2 v0.20.5	Two views; 16 tokens; label only	256 / 256	FIFO; S1 B1; shorter Safety contract
E3 v0.20.8	Two views; 16 tokens; label only	32 / 16	Parallel roles, phases, MPS; S1 B2 with 2 ms formation wait; no geometry or retreat
E4 v0.20.10	Overview only; 64 tokens; visibility, person, collision	32 / 16	E3 colocation plus geometric prediction; no independent retreat
E5 v0.20.11	Qwen3 4B/8B; two views; 16 tokens; label only	32 / 16	No geometry or retreat; Safety memory fraction 0.15 for 4B and 0.25 for 8B
E6 v0.20.12	Overview only; 64 tokens; visual guard	32 / 16	Geometry, independent retreat and replanning; .13 is audit correction only
E7 v0.20.14	Overview only; 64 tokens; visual guard	32 / 16	Feedback limited to S2; Safety concurrency 2; hard-reset preparation
E7 v0.20.15	Same as v0.20.14	32 / 16	Adds MPS priority separation: S1/Safety 0, S2/Monitor 1
E7 v0.20.16	Same as v0.20.15	32 / 16	Changes recording to local disk followed by verified persistent copy

Final model input contract

Saved images are 256 × 256 RGB. The final VLM settings use seed 42, FP16, maximum sequence length 4,096, min_pixels=50,176, max_pixels=200,704, prefix caching disabled and TORCH_SDPA vision attention. S2 returns a short subgoal; Safety returns visibility, person and collision; Monitor returns status. Generation limits and view counts in the final configuration must not be retroactively applied to older trials.

S2 reuse is consumed by every S1 call, including failed or unapplied calls, and resets at episode change. Monitor records ongoing, done, failed or unknown. With geometry enabled, collision ambiguity may be delegated to MuJoCo; person detection and unknown visibility are not reclassified as safe. Full prompts, parsers and per-series settings are preserved in the matching source archives.

The Qwen3 comparison also changed model-specific tokenizer, image processor and chat template, and the 8B memory reservation. It is not a weights-only comparison. Fixed model revisions appear in Table B2.

Appendix D Initial states and experiment identities

Table D2. Closed loop cases and completed measurement windows

Case ID	Initial states / seeds	Warmup / measurement	Measured replicates
E1-P1	0-49 / 42-91	15 s / 30 s	1
E1-P234	0-49 / 42-91	60 s / 120 s	1
E2-FIFO	0-49 / 42-91	60 s / 120 s	1
E3	0-49 / 42-91	60 s / 300 s	3 fixed
E4	20-49 / 62-91	60 s / 300 s	1 of 3 planned
E5-4B	0-9 / 42-51	60 s / 120 s	1
E5-8B	0-9 / 42-51	60 s / 120 s	1
E6	0-9 / 42-51	60 s / 300 s	1 exploratory
E7-v14	0-9 / 42-51	60 s / 300 s	1 exploratory
E7-v15	20-49 / 62-91	60 s / 300 s	3 fixed
E7-v16	20-49 / 62-91	60 s / 300 s	2 of 3 planned

Indices identify original T01 states, not reached episodes. Start positions shift by robot index within class and replicate index, then cycle with original seeds. E1-P1 uses four environments and B4; other cases use one each for P2/P3/P4. E3 onward uses parallel roles, phases, MPS and S1 B2 with a 2 ms formation wait.

E4 ran only its first unsuccessful confirmation. E7-v16 trial 2 finished measurement but stopped during persistent copying, leaving its audit incomplete. The CPU-only v0.20.17 fix was not applied to these results.

Table D3. Evidence groups and preserved source versions

ID	Source versions	Scope
E1	v0.20.1 / .2 / .3	Numerical and behavioral regression; P1 and initial P2-P4; early profile correction
E2	v0.20.4 / .5	Recording separation, individual profiles, Safety shortening and FIFO/deadline-order comparison
E3	v0.20.8	Fixed timing confirmation
E4	v0.20.10	MuJoCo prediction and rejection analysis
E5	v0.20.11-4b / -8b	Separate corrected sources for Qwen3 4B and 8B
E6	v0.20.12 / .13-audit	Retreat/replanning; .13 corrects recovery aggregation boundaries
E7	v0.20.14 / .15 / .16 / .17-cpu	Visibility, timing and confirmation; interrupted recording; CPU-only termination fix

Appendix E Complete stop results and latency diagnostics

Table E1. Original S1 stops in the reported measurement windows

Series and condition	P3 stops / decisions	P4 stops / decisions	Recorded qualification
MuJoCo only 300 s [E4]	157/157 = 100%	85/210 = 40.5%	P3 exceeded the 90% stop-rate target; P4 met it
Qwen3 4B 120 s [E5]	238/238 = 100%	205/205 = 100%	Before geometric checking and retreat
Qwen3 8B 120 s [E5]	239/239 = 100%	204/204 = 100%	P4 applied three actions in warmup; none in measurement
Retreat added 300 s [E6]	450/513 = 87.72%	2/242 = 0.83%	P3 stopped after one resumed chunk
v0.20.15 trial 1 [E7]	21/286 = 7.34%	0/253 = 0%	P3/P4 S1 p99 and P3 Monitor period failed
v0.20.15 trial 2 [E7]	15/288 = 5.21%	0/257 = 0%	P3 Monitor period failed
v0.20.15 trial 3 [E7]	16/285 = 5.61%	1/253 = 0.40%	All fixed criteria passed
v0.20.16 trial 1 [E7]	14/294 = 4.76%	524/524 = 100%	Timing passed; P4 recovery failed

Version v0.20.16 trial 2 is excluded from finalized stop-rate results. Its complete recording was retained separately; P3 Monitor fulfilled 148/150 slots, or 98.67%. No third trial was started. Trials differ in inputs, execution and trajectories; these rates are not paired comparisons.

Table E2. Measured latency and queue diagnostics

Diagnostic	Observed result
FIFO after Safety shortening [E2]	Mean S1 model call 63.19 ms versus mean queue wait 2,644.95 ms. Mean Safety model call 104.00 ms versus mean wait 2,783.77 ms.
Safety contract comparison [E2]	On the same 10 inputs, mean latency changed from 3,099.44 to 104.91 ms. Three decisions changed: two unsafe to unknown, one safe to unknown.
Deadline-ordered queue [E2]	81 normal responses, 305 pre-inference deadline rejections and 8 cancellations. Warmup was insufficient; measurement did not start and Q was not measured.
Prediction restoration [E4]	Of 157 P3 stops, 18 exceeded the 2,000 ms restoration limit and 139 were rejected by the fingertip-to-basket distance condition.

Safety shortening changed the prompt and output schema as well as the token limit from 256 to 16. The timing difference cannot be attributed to token count alone. The S1 8 Hz profile used the 227-observation manifest and its fixed partitions; it is separate from the closed loop windows in Table D2.

Appendix E Diagnostic evidence and unresolved questions

Table E3. Recorded findings from recovery and termination analysis

Issue	Confirmed evidence
Unknown visibility after retreat [E7]	On the same saved image, removing added replanning feedback changed unknown to clear. A completely black image remained unknown.
Visibility after the correction [E7]	The v0.20.16 exploration and confirmation trials 1 and 2 each had zero P3 unknown-visibility outputs among 600 decisions.
P4 recovery failure [E7]	Trial 1 predicted approximately 1.752 mm between fingertip and floor, below the 2 mm threshold. The consecutive-retreat limit was reached; original S1 application count was zero in measurement.
Missing terminal Monitor requests [E7]	CPU regression reproduced an observation-save delay crossing scheduled dispatch, followed by termination cancelling an already scheduled request. The fix passed 45 CPU tests; GPU validation was not performed.
Rejected alternatives [E7]	Soft reset was not adopted because continuation-state equivalence was unconfirmed. A three-character Safety output was rejected after person/collision labels became invalid.

Interpretation boundaries

The same-image diagnostic restricted input changes; whole trials did not hold trajectories constant. P4 stopping after the v0.20.16 recording change does not identify storage as its cause. GPU interference, host scheduling, rendering and reset effects remain unisolated.

The dominant P4 cause remains unknown. Before changing the 2 mm criterion, verify allowed and prohibited contacts, coordinate conventions and action scale. Evaluate the threshold separately: fewer rejections do not prove safe continuation.

A candidate recovery study should freeze failed states, compare multiple retreat directions, local path planning, regrasping and reinitialization, and track object retention, reachability, collisions, rejection and eventual task success. Timing work should separately measure storage I/O, IPC, reset, queueing, inference and prediction, using the same input distribution and load.

Semantic evaluation should include human labels, negative examples for people, occlusion and contact, and multiple raters for ambiguous cases. Compare shortened contracts and model replacements on the same evaluation set. Complete failure handling also requires bounded task retry, stop or handoff, and component-specific fallback without duplicate actions or stale-response reuse. These are proposed tests, not completed results.

Evidence not yet obtained

Accepted full profiles for GPU optimization, human-labeled Safety/Monitor accuracy, full R06 behavioral equivalence, multi-GPU comparisons and monetary/energy savings remain unmeasured. The paper's maximum speedup and capacity were not reproduced. Ninety saved evaluation forms support future semantic assessment; recorded model outputs alone do not measure accuracy.

Appendix F Reproduction materials and traceability

Use the unchanged Revision 1 companion archive, ROSA_reproduction_reference_20261101_rev1.zip. Source archives are preserved byte for byte; exported settings use deployment variables. Original and rebuilt suite identities remain distinct. No public repository URL has been issued.

SHA-256: ee83463004ccd24c7aea2abbaa5baf974f280b580e67ca4545e8013085046f5d

Table F1. Companion archive inventory

Location	Contents and purpose
reproduction_index.json	E1-E7 conditions, versions, original suites and exclusions
sources/	15 archives; index.json stores archive and runtime hashes
experiments/	11 closed loop conditions, four E2 individual measurement conditions, model settings and state banks
inputs/	227 JSON/NPZ observations; corpus.json; split 105/32/90
environment/	Recorded shared_environment_lock.json for GPU execution
prepare_reproduction.py paths.template.json	Rebuilds plans and suites for supplied paths; does not launch Ray or GPU models
README.txt SHA256SUMS	Environment and entry-point guidance plus hashes for distributed files

CPU preparation validated all 11 cases against saved source, state banks and runtime hashes; it adds no GPU results. Model weights, credentials and complete time-series logs are excluded. Recomputing all historical aggregates requires the original logs. The GPU lock differs from CPU-only requirements.lock.

Table F2. Execution entry points and evidence to retain

Step	Entry points	Evidence
Environment	shared_environment.py; groot_checkpoint.py	Dependency lock, complete weights, CUDA/EGL checks and runtime identity
Inputs	observations.py	Fixed states, seeds and observation manifest
Regression	r02_compare.py; r02_t01.py	Official API versus adapter at B1/2/4/8/16
Individual profiles	postpod_models.py; runtime_evidence.py	Scheduled requests, responses and CUDA/kernel evidence
Closed loop	prototype_validation.py; pipeline_experiments.py	Fixed suite, actions, periodic events and GPU monitoring
Confirmation and collection	Version-specific audits and collection tooling	Retain failed trials; distinguish interrupted audits; verify hashes

Appendix F Deployment and execution procedure

1. Extract the companion archive and verify SHA256SUMS. Select a case in Table D2 and match its source archive using sources/index.json. Acquire LIBERO, Isaac-GR00T and model weights at the fixed revisions in Table B2. Do not place authentication tokens in configuration files or logs.

2. Prepare Linux Python 3.12 environments for common model execution and simulation. Use environment/shared_environment_lock.json and the selected version's shared_environment.py, docs/shared-formal-comparison.md and docs/locked-dynamic-mixed.md. A CPU requirements.lock is not a full GPU environment lock. Recheck complete weights, CUDA, EGL and model input/output contracts; historical runtime_fingerprint or stateless_verified values do not certify a new environment.

3. Copy paths.template.json to paths.json and supply deployment paths. With the preparation dependencies installed, run the following from the extracted archive root. The helper extracts source and state banks, rebuilds plan/protocol/suite using that exact version and validates them without starting Ray or models.

```
python prepare_reproduction.py --case E7-v16 \
  --paths paths.json --output prepared/E7-v16
```

Path changes produce a new suite_id; preparation.json retains the original identity. The helper preserves planned repetitions, including those not executed historically. Do not count newly run repetitions as past evidence.

4. Start Ray in the common environment. Set MUJOCO_GL=egl and PYOPENGL_PLATFORM=egl; fix OMP_NUM_THREADS, OPENBLAS_NUM_THREADS and MKL_NUM_THREADS to 8. For MPS series, configure the same pipe/log settings and client priorities before model startup. Define TRIAL_ROOT as the absolute preparation directory; define COMMON_PYTHON, SIM_PYTHON, GROOT_REPO and RAY_ADDRESS for the deployment.

```
"$COMMON_PYTHON" "$TRIAL_ROOT/source/scripts/prototype_validation.py" run \
  --suite "$TRIAL_ROOT/suite.json" --root "$TRIAL_ROOT" \
  --output "$TRIAL_ROOT/new-run-1" --sim-python "$SIM_PYTHON" \
  --sim-repo "$GROOT_REPO" --address "$RAY_ADDRESS" --rep 0
```

5. Replicate indices start at zero. Use new model instances and a fresh output directory for every run. Record GPU/MPS monitoring in parallel. Evaluate original S1 stops, component p99, periodic fulfillment and recovery separately: a CLI status of complete denotes execution completion, not acceptance of all criteria. For v0.20.16, record on local host storage and copy to persistent storage with hash verification afterward.

Individual profiles and Safety comparisons

For E2, replace deployment variables in E2-probes/*.template.json and use v0.20.4 postpod_models.py prepare to rebuild the suite. The S1 profile uses stage=profile, phase=formal, rates=8 and repetitions=3. Acquire fresh smoke and kernel evidence first; pass the new smoke record via --smoke-evidence. Use inputs/corpus.json and inputs/capture. Keep the same 10 inputs for baseline and shortened Safety contracts. The archive README provides additional details; this English procedure supplements it.

Appendix G References

This English submission reorganizes Revision 1 using experiment records through 1 November 2026, without adding GPU results. Public references were checked for Revision 1 on that date. E1-E7 are local records mapped in Table D3 and the companion archive, not third-party publications. Figure 1 is an original schematic of this prototype.

[1] Wenqi Jiang et al. ROSA: A Robotics Foundation Model Serving System for Robot Factories. arXiv:2607.01088v1, 2026. https://arxiv.org/abs/2607.01088v1

[2] Mustafa Shukor et al. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics. arXiv:2506.01844v1, 2025. https://arxiv.org/abs/2506.01844v1

[3] Moo Jin Kim, Chelsea Finn and Percy Liang. Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success. RSS 2025; arXiv:2502.19645v2. https://arxiv.org/abs/2502.19645v2

[4] Woosuk Kwon et al. Efficient Memory Management for Large Language Model Serving with PagedAttention. SOSP 2023; arXiv:2309.06180. https://arxiv.org/abs/2309.06180

[5] Kaiyuan Chen et al. FogROS2-Config: Optimizing Latency and Cost for Multi-Cloud Robot Applications. ICRA 2024; arXiv:2311.05600v2. https://arxiv.org/abs/2311.05600v2

[6] NVIDIA. GR00T-N1.7-LIBERO model card. https://huggingface.co/nvidia/GR00T-N1.7-LIBERO

Isaac-GR00T source, fixed commit in Table B2. https://github.com/NVIDIA/Isaac-GR00T

[7] Qwen Team. Qwen2.5-VL model cards. https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct

Safety model card. https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct

[8] NVIDIA. Multi-Process Service, Appendix: Environment Variables. https://docs.nvidia.com/deploy/mps/appendix-environment-variables.html

[9] Ray Team. Dynamic Request Batching. Ray 2.59.0 documentation. https://docs.ray.io/en/latest/serve/advanced-guides/dyn-req-batch.html

[10] LIBERO Project. Benchmark and source. https://libero-project.github.io/

LIBERO source, fixed commit in Table B2. https://github.com/Lifelong-Robot-Learning/LIBERO

[11] Ray Team. Resources: Physical Resources and Logical Resources. Ray 2.59.0 documentation. https://docs.ray.io/en/latest/ray-core/scheduling/resources.html#physical-resources-and-logical-resources